

BUCIN, Romania

WMO: 151480

Lat: 46.6489N

Lon: 25.2964E

Elev: 1284

StdP: 86.82

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.1	(c) -14.5	(d) -20.3	(e) 0.7	(f) -15.3	(g) -18.2	(h) 0.9	(i) -12.4	(j) 7.4	(k) -8.8	(l) 6.2	(m) -6.9	(n) 2.0	(o) 50	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.4	(c) 24.0	(d) 16.4	(e) 22.3	(f) 15.5	(g) 21.0	(h) 14.8	(i) 17.1	(j) 22.1	(k) 16.3	(l) 20.9	(m) 15.6	(n) 19.8	(o) 2.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	12.9	18.1	14.7	12.2	17.8	13.9	11.6	17.5	53.3	22.2	50.7	21.0	48.5	20.0	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.0	(b) 5.0	(c) 4.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 6.0	(b) 5.0	(c) 4.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.6	-3.6	-1.4	4.2	9.8	13.5	14.3	15.1	10.4	5.7	2.0	-3.8
	DBStd	8.46	5.02	4.63	4.56	5.24	3.49	3.25	3.12	3.36	4.93	4.69	4.75	4.66
	HDD10.0	2339	485	381	354	184	46	8	3	4	59	148	240	428
	HDD18.3	4853	743	614	612	426	264	148	128	108	241	393	489	687
	CDD10.0	544	0	0	0	8	39	112	138	162	70	14	1	0
	CDD18.3	16	0	0	0	0	0	2	4	7	2	0	0	0
	CDH23.3	75	0	0	0	0	0	10	16	39	10	0	0	0
	CDH26.7	2	0	0	0	0	0	1	0	2	0	0	0	0
Wind	WSAvg	1.6	1.6	1.8	1.9	1.8	1.6	1.6	1.6	1.6	1.6	1.6	1.4	1.7
Precipitation	PrecAvg	720	31	34	42	62	82	101	95	70	62	56	42	49
	PrecMax	960	56	75	105	109	141	179	162	120	147	112	96	157
	PrecMin	517	12	7	6	24	24	26	34	22	6	1	0	12
	PrecStd	110	15	17	24	25	30	36	37	26	35	30	21	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.8	12.7	18.7	21.2	24.4	25.1	26.4	25.1	19.6	14.2	7.8
		MCWB	2.3	4.0	6.3	10.3	12.3	16.5	16.8	16.8	16.9	11.6	7.6	2.9
	2%	DB	3.6	6.0	9.1	16.5	19.1	22.2	23.2	24.2	21.5	17.3	12.3	5.7
		MCWB	1.5	2.8	4.0	9.3	12.1	15.8	16.4	16.3	14.5	10.9	6.9	2.6
	5%	DB	2.2	4.2	6.8	14.2	17.4	20.8	21.5	22.4	19.4	14.8	10.4	3.9
		MCWB	0.8	1.6	2.8	8.5	11.7	15.2	15.5	15.5	13.4	9.9	6.3	1.7
	10%	DB	0.6	2.5	4.6	11.7	15.6	19.1	19.9	20.8	17.6	12.3	8.5	2.3
		MCWB	-0.5	0.9	1.8	7.1	10.9	14.2	14.6	15.0	12.7	9.0	5.3	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	5.6	6.9	10.9	14.0	18.1	17.6	18.2	17.0	13.1	9.2	4.8
		MCDB	4.6	7.1	13.0	16.9	18.8	22.0	22.7	24.0	24.7	17.4	12.9	6.3
	2%	WB	2.2	3.1	4.8	10.0	12.9	16.6	16.8	17.1	15.1	11.6	8.0	3.5
		MCDB	3.3	4.7	8.2	15.2	17.0	20.6	21.7	22.1	20.2	15.5	10.7	4.8
	5%	WB	1.0	2.1	3.5	9.1	12.2	15.7	16.1	16.2	13.8	10.5	7.0	2.3
		MCDB	2.0	3.7	6.4	13.8	16.3	19.6	20.3	21.0	18.3	14.4	9.5	3.7
	10%	WB	-0.4	1.1	2.2	7.6	11.3	14.9	15.3	15.6	12.9	9.1	5.9	0.9
		MCDB	0.5	2.3	4.2	11.3	15.1	18.6	19.0	20.0	17.0	11.9	8.0	2.0
Mean Daily Temperature Range	5% DB	MDBR	5.7	6.3	6.9	8.0	7.9	8.3	8.0	8.4	7.2	7.4	5.9	5.6
		MCDBR	6.8	8.0	9.9	11.2	10.2	10.3	10.0	10.4	10.4	11.1	8.7	6.9
		MCWBR	5.0	5.6	6.2	6.1	5.5	5.3	5.4	5.2	5.2	6.7	5.6	4.6
	5% WB	MCDBR	5.9	6.5	8.7	10.1	9.5	9.5	9.1	9.4	9.2	10.2	7.4	6.2
		MCWBR	4.8	5.1	5.6	5.8	5.5	5.4	5.3	5.2	5.2	6.4	5.1	4.6
Clear-Sky Solar Irradiance	taub	0.272	0.298	0.337	0.385	0.367	0.364	0.382	0.387	0.351	0.320	0.299	0.275	
	taud	2.372	2.330	2.311	2.219	2.329	2.359	2.307	2.291	2.372	2.458	2.464	2.418	
	Ebn at Noon	829	873	885	868	898	900	877	857	860	840	792	784	
	Edh at Noon	71	91	109	132	122	119	124	121	102	80	66	61	
All-Sky Solar Radiation	RadAvg	1.28	2.13	3.24	4.33	5.36	5.79	5.73	5.28	3.75	2.57	1.44	0.97	
	RadStd	0.17	0.23	0.38	0.43	0.49	0.48	0.49	0.47	0.43	0.32	0.19	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+0.70	N/A	N/A	+0.85	N/A	N/A	-153	-215	+87	+23

Nomenclature: See separate page